ChatGPT

Physical Layout Strategy for a Fabricated SilicaFlux ASIC

Executive summary

SilicaFlux is physically much closer to a **small tiled accelerator SoC** than to a conventional microcontroller. Direct inspection of the uploaded RTL shows a scalar control pipeline feeding a scheduler, four compute tiles arranged naturally as a 2×2 fabric, a 2×2 NoC, per-tile scratchpad memories, a shared scalar-side memory controller, an instruction memory, a scalar/vector register bank and a narrow chiplet-facing serial interface. The uploaded assessment independently identifies the project as a roughly 30-kLOC, 176-module SystemVerilog codebase containing CPU, DSP, cache/memory and SoC functions, while also noting that verification is not yet at production-ASIC closure level. 🗄filecite🗄turn0file0🗄

The most important physical-design conclusion is that **the RTL hierarchy already suggests the floorplan**: the four `compute_tile` instances should become four nearly identical physical islands surrounding a central scheduler/control spine, with their local SRAM macros placed on the inward-facing edge of each tile and the NoC routers at the tile corners nearest the centre. This minimises the highest-activity interconnect lengths, gives the NoC a geometrically natural 2×2 topology, and keeps each tile's vector/tensor datapath close to its scratchpad. The instruction SRAM and scalar register/memory complex should sit adjacent to the scalar control core rather than at a die edge.

For a first serious ASIC, **65 nm is the best overall implementation target**. At SilicaFlux's current 100 MHz timing constraint, 180/130 nm is entirely feasible but becomes unnecessarily large and pad-dominated; 28 nm offers materially better energy and density but raises NRE, sign-off complexity and implementation risk before the RTL and verification environment have matured. A 7 nm implementation is technically possible but economically irrational for the present design size and performance target: TSMC states that N7 provides roughly 3× the logic density of N16, with substantial power and speed benefits, but that level of scaling is valuable only when the design needs the density or frequency. [1]

For first silicon I would therefore use a **65 nm standard-cell flow, compiled SRAM macros, foundry PLL and I/O cells, one voltage domain initially, multi-Vt optimisation, a hierarchical buffered clock tree rather than a mesh, and a regular top-metal power grid**. The four compute tiles should be placed as repeated rectangular hard regions, not allowed to disperse globally during placement. A later 28 nm revision could introduce multiple voltage domains and more aggressive power gating.

A representative 65 nm physical target is approximately **0.8–1.5 mm² of active core logic and memories before utilisation margin**, expanding to perhaps **1.3–2.2 mm² of core floorplan after routing, clock, power and whitespace**, while the final die is likely to be **I/O-pad limited**, plausibly around 2–3 mm on a side depending on pad count and package. These figures are engineering estimates, not post-synthesis measurements; an actual area must come from synthesis against a selected library and SRAM compiler.

The single biggest tapeout risk is not floorplanning. It is **design maturity**. The current project has FPGA-oriented SDC constraints, simulation-centric tests, and no evidence yet of full production STA, CDC/RDC closure, scan insertion, MBIST, ATPG, IR-drop analysis, EM sign-off or foundry-qualified SRAM/PLL integration. The earlier assessment similarly finds good RTL modularity but incomplete industrial verification infrastructure. filecite turn0file0 A physical design should not begin as an optimisation exercise until those items have a defined closure plan.

Mapping the RTL into physical macros

Direct inspection of `silicaflux_pkg.sv` gives the key architectural constants: 32-bit scalar datapath, 128-bit vectors, 32 scalar registers, 16 vector registers, four compute tiles, four SIMD lanes per tile by default, 256×32-bit local scratchpads, a 2×2 NoC and a 1024×32-bit instruction memory. `compute_tile.sv` instantiates one vector engine, one tensor unit and one memory controller per tile. The tensor engine is deliberately sequential rather than a 16-MAC fully parallel array, which matters substantially for physical area and power.

The storage implied by the default RTL is modest. Four tile SRAMs provide 4 × 1 KiB = 4 KiB; the scalar-side `memory_controller` adds another 1 KiB; the 1024×32 instruction SRAM adds 4 KiB. The architectural register files and local vector registers add less than another kilobyte. The design therefore contains only about **10 KiB of architecturally obvious on-chip storage**, meaning memory macros will matter physically but will not dominate the die as they would in a cache-heavy processor.

A sensible 65 nm macro budget is:

Physical block	RTL source / role	Recommended implementation	Indicative 65 nm active area	Preferred aspect ratio	Placement constraint
Scalar control complex	`fetch_unit`, `decoder`, `control_core`, ALU, forwarding	Standard cells	0.08–0.16 mm²	1.0–1.5	Close to IMEM, register file and scheduler
Instruction SRAM	`IMEM_DEPTH=1024`, 32-bit	Compiled SRAM macro	0.03–0.08 mm²	Compiler dependent, often tall rectangle	Immediately adjacent to fetch stage
Scalar scratchpad	shared `memory_controller`	Compiled 1 KiB SRAM	0.01–0.03 mm²	Compiler dependent	Near scalar MEM stage
Register bank	32×32 + 16×128	Flops for first spin or small RF macro	0.02–0.05 mm²	1–2	Between decoder/control and scheduler
Scheduler	scoreboard, tile dispatch, barrier	Standard cells	0.02–0.05 mm²	~1	Exact centre of four tiles
One vector engine	four 32-bit lanes, including multiplies	Standard cells / multiplier cells	0.04–0.09 mm²	1–2	Beside tile SRAM

Physical block	RTL source / role	Recommended implementation	Indicative 65 nm active area	Preferred aspect ratio	Placement constraint
One tensor engine	sequential 32×32 multiply-accumulate FSM	Standard cells	0.03–0.07 mm²	1–1.5	Beside vector engine
One tile SRAM	256×32 = 1 KiB	Compiled SRAM	0.01–0.03 mm²	Compiler dependent	Inner edge of each tile
One NoC router	five-port local XY router	Standard cells	0.01–0.025 mm²	~1	Tile corner facing chip centre
Complete compute tile	vector + tensor + SRAM + router/control	Hierarchical placed block	0.11–0.20 mm²	0.8–1.25	Fixed quadrant
Four-tile fabric	four repeated tiles	Hierarchical standard-cell blocks	0.45–0.80 mm²	~1	Symmetric 2×2
Chiplet/tile interface	serializer, 2-FF retiming	Standard cells + hard PHY if truly off-chip	0.02–0.05 mm² excluding pads	edge-oriented	Die edge
PLL	absent from present RTL	Foundry hard macro	0.02–0.10 mm² node dependent	fixed macro	Quiet corner, dedicated supply
I/O / ESD ring	top-level external signals	Foundry I/O cells	pad-limited	ring	Die perimeter
Scan/ MBIST/test	not currently architected	DFT logic	+8–20% logic	distributed	Scan chains local to physical partitions

Those are planning ranges rather than library-characterised results. The right next quantitative step is synthesis against one real technology library and SRAM compiler, because RTL line counts or FPGA LUT counts do not translate reliably into silicon area.

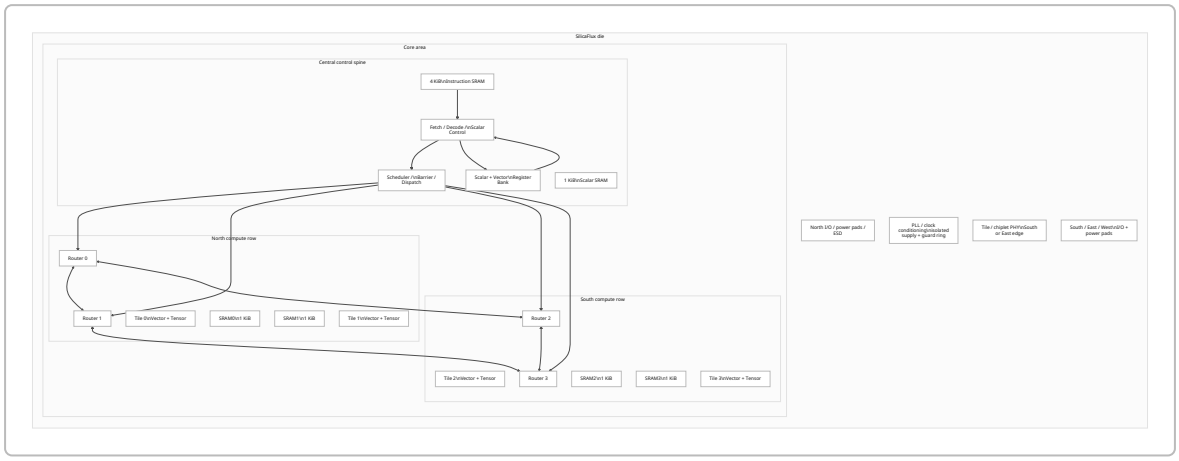
The tensor unit deserves particular attention. Its source stores sixteen A elements, sixteen B elements and sixteen accumulators but performs multiplication sequentially through a single wide multiplier datapath. Physically it is therefore much smaller than the words “4×4 tensor unit” might suggest. Conversely, the vector engine can instantiate a multiplication function per lane, so **the VMUL lanes may contribute more instantaneous switching current than the sequential tensor block**. That is counter-intuitive but important for IR-drop and hotspot modelling.

The earlier assessment's simplified SoC diagram places CPU/control, memory and peripherals around a bus. The current SilicaFlux accelerator archive is more naturally interpreted as a four-tile compute

cluster; its repeated tile structure should be preserved physically rather than flattened.
filecitefile0

Recommended floorplan

The preferred floorplan is a **square or near-square core with four symmetric compute quadrants** and a narrow control/memory spine running through the centre. The scheduler sits at the geometric centre because it fans out to every tile. The scalar core, register bank and instruction SRAM occupy one side of that spine. The NoC routers occupy the four inner corners of the compute tiles.



The critical placement principle is that **memory is local to compute**. Each tile SRAM should face inward toward its vector/tensor datapath rather than outward toward the pad ring. The SRAM data bus is 32 or 128 bits depending on operation, and repeated local accesses will dominate routing activity. Shortening those buses reduces both dynamic power and routing congestion.

The four routers should form a true geometric 2×2 mesh. Physical placement that mirrors logical topology is particularly valuable for NoCs because it makes the east-west and north-south links short, uniform and timing-predictable. The central scheduler should not directly stretch long wires to datapaths across the die; its dispatch signals should terminate in tile-local control, after which local logic drives the compute engines.

The I/O ring surrounds the complete die. Core-facing pad cells should be arranged by interface rather than arbitrary pin order. The narrow chiplet-facing interface should occupy one contiguous edge sector. Reset, JTAG/test and low-speed control can occupy another sector. Power and ground pads should be distributed around all four sides rather than concentrated near one corner.

There is **no analogue or RF subsystem in the present RTL**. A PLL would therefore be added as foundry IP, not synthesised from SystemVerilog. It should be placed near a die corner or edge, physically separated from vector multipliers and clock-gating bursts. Its analogue supply should have dedicated pad access, local decoupling, substrate guard rings and routing keep-outs. At GF180, the public PDK explicitly provides guard-ring options and analogue-layout guidance; foundry guidance also warns about sensitive analogue circuitry near certain structures. 2

For an advanced-node version, the PLL and any high-speed chiplet PHY should be treated as hard IP with explicit routing blockages over sensitive analogue regions. Their reference clocks and differential inputs should not cross the compute fabric.

OpenROAD's hierarchical macro placer explicitly supports macro guidance regions, minimum channels and target utilisation, which reflects standard physical-design practice: large memories should be fenced into architecturally appropriate regions rather than left entirely unconstrained. [3]

A good initial core utilisation is **55–60%**, not 70–80%. That deliberately leaves whitespace for SRAM access routing, top-level NoC channels, CTS buffers, decaps and post-route optimisation. The design is small enough that chasing aggressive utilisation on first silicon offers little financial benefit.

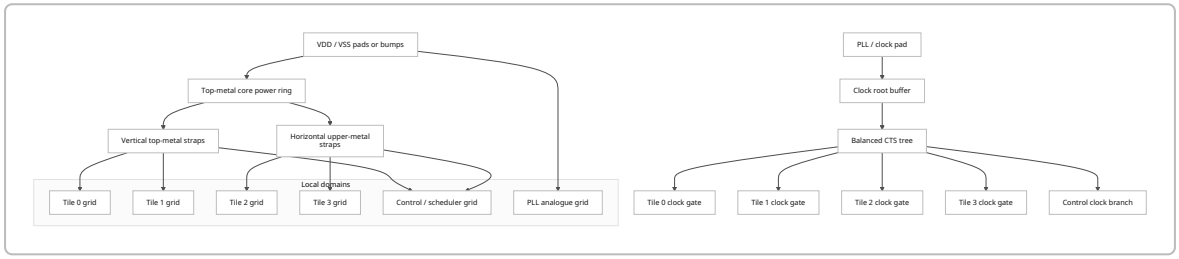
Cell architecture, power delivery and clocks

For SilicaFlux, almost all digital logic should be implemented using conventional foundry-characterised standard cells. Full-custom layout would make sense only for a future highly replicated MAC block or a custom SRAM; neither is justified at current scale.

Block type	Standard-cell	Gate-array / structured ASIC	Full-custom	Recommended choice
Scalar pipeline	Excellent flexibility, easy ECO	Possible but little advantage	Excessive NRE	Standard cell
Scheduler / NoC	Excellent	Possible	Poor value	Standard cell
Vector ALUs	Good, synthesis-friendly	Possible	Useful only at extreme PPA targets	Standard cell initially
Multipliers	Library/datapath synthesis	Efficient in some structured fabrics	Potential PPA advantage	Standard-cell/ datapath cells
Tensor engine	Fits standard flow well	Possible	Not justified for sequential design	Standard cell
Register file	Flops or small memory macro	Possible	Custom only if repeated/large	Flops or compiler macro
Scratchpads / IMEM	Inferable RTL but poor if implemented as flops	Not ideal	Custom SRAM expensive	Foundry SRAM compiler
PLL	Impossible as ordinary synthesised digital logic	No	Analogue custom/hard IP	Foundry hard macro
I/O / ESD	No	No	Foundry-specific	Foundry hard cells
Chiplet PHY	Digital wrapper standard cells	No	High-speed PHY normally hard IP	Hard PHY if > low-speed GPIO

The **SRAM conversion is mandatory**. The RTL arrays are suitable for FPGA inference, but an ASIC implementation should replace them with wrapper modules around foundry memory-compiler instances. Leaving a 4 KiB instruction memory or multiple 1 KiB scratchpads as flip-flops would waste area and power.

For power distribution, use a hierarchical grid: package bumps or power pads feed top-metal rings, wide top-metal horizontal and vertical straps cross the core, intermediate metals distribute into each tile, and low metal connects cell rails. OpenROAD's PDN tooling expresses exactly this distinction between standard-cell grids, macro grids and voltage domains. [4]



For the first ASIC I would use **one digital VDD domain**. The architecture does support a very attractive future partition: one always-on control/scheduler domain and four independently gateable tile domains. But adding multiple supplies before basic functionality and scan/MBIST are stable introduces level shifters, isolation, retention decisions, power-state verification and UPF complexity that are unnecessary for first silicon.

Multi-Vt, however, is worthwhile immediately. Use low-Vt cells only on the handful of paths that fail timing; use regular-Vt broadly; use high-Vt cells on non-critical control and NoC state to reduce leakage. This achieves much of the leakage benefit of a more elaborate power architecture without introducing multiple rails.

IR-drop analysis should be activity-driven. Run vector VMUL, DOT and simultaneous four-tile traffic as worst-case dynamic scenarios. Analyse both static and dynamic voltage drop after placement and again after route. The key question is not average chip current but **simultaneous switching in the four vector engines plus SRAM read/write edges**. Extra straps and decaps should be concentrated around the tile datapaths.

The GF180 public PDK illustrates why this must be sign-off driven rather than based on generic width rules: it provides multiple metal-stack options, explicit metal density constraints, via rules and current-density guidance. It also recommends avoiding single vias where possible for reliability. [5]

For clocks, the present SDC requests only **100 MHz with a 10 ns period**. At that frequency and at this die size, a clock mesh is unnecessary. A balanced buffered tree, optionally with an H-tree-like top level, provides lower clock power and adequate skew.

OpenROAD CTS supports sink clustering, macro clustering, obstruction awareness, non-default routing and useful skew; its placement flow can also model virtual clock latency before final CTS. [6]

Recommended first-pass targets at 65 nm are approximately:

Clock metric	100 MHz SilicaFlux target
Clock period	10 ns
Post-CTS global skew	<100 ps preferred; <200 ps acceptable
Clock insertion delay	<0.5–1.0 ns
Clock transition	library-dependent, typically comfortably below 10% of period
Clock uncertainty budget	0.2–0.4 ns initial
Clock-gating granularity	each compute tile + potentially vector/tensor sub-blocks

The RTL already has natural idle boundaries around scheduler/tile execution. Those should become integrated-clock-gating enables rather than RTL-generated clocks. Never create combinationally gated clocks.

Clock mesh or tree-plus-grid distribution becomes interesting only if a later implementation runs the accelerator at several hundred MHz to >1 GHz and if local variation dominates the skew budget. Historically, tree-plus-grid architectures have demonstrated very low skew on large high-performance processors, but at a significant capacitance and power cost. [7]

Routing, sign-off, thermal and manufacturability

The highest-priority routing resources are the global clock, power straps, 128-bit vector buses, NoC links, SRAM buses, scheduler dispatch signals and any future high-speed PHY. The physical strategy should reserve upper metals for global traffic and keep lower metals local.

A representative layer policy is:

Layer class	Preferred use
M1–M2	Standard-cell local connections and pins
M3–M4	Tile-local data buses, SRAM interfaces
Mid-upper metals	NoC links, long scheduler/control nets
Top two routing metals	Power grid and clock trunks
Thick top metal where available	Power ring, package connections, high-current nets

On a five-metal 130/180 nm technology there is much less freedom. SKY130, for example, provides five metal levels. [8] A GF180 implementation can be ordered with several BEOL configurations, up to six routing metals in the public process documentation. [9] This is one reason a 65 nm technology with a richer metal stack is attractive for a tiled accelerator even if logic density itself is not required.

NoC links should use repeated, predictable layer assignments. In a square 2×2 floorplan, east-west channels should preferentially use one metal direction and north-south channels the orthogonal direction. That reduces jogs and makes crosstalk easier to reason about.

Critical clock and reset trunks should receive non-default rules where available: wider wires, larger spacing, redundant vias. Power-grid vias should use arrays rather than single cuts. GF's public 180 nm rules explicitly recommend multiple vias for yield and reliability. [10]

There are no RTL-defined differential signals. If a high-speed chiplet PHY is introduced, then reference clock and serial differential pairs become hard-PHY concerns and must follow the PHY vendor's matched-length, impedance, shielding and keep-out requirements.

The advanced-node interconnect problem becomes increasingly severe. IRDS identifies rising resistance, electromigration limits, dielectric reliability and a widening conflict between local signal routing and low-resistance power delivery as major scaling challenges. [11] The IRDS interconnect roadmap also highlights line/via roughness, integration complexity and local power-routing congestion as fundamental issues. [12] That is another reason 7 nm provides little benefit to a modest 100 MHz accelerator: the physical implementation becomes more difficult while the architecture cannot exploit much of the available transistor performance.

Thermally, likely hotspots are the vector multipliers rather than the memories. The tensor unit is sequential, so it spreads work over time; four simultaneous vector VMUL or DOT operations are a more plausible peak-power event. Keep the four compute blocks distributed around the centre instead of clustering all arithmetic on one side. This creates thermal symmetry and prevents a single concentrated hotspot.

Reliability closure must include EM, IR-drop, antenna, ESD, latch-up, ageing and thermal analysis. At advanced nodes, IRDS notes that interconnect resistance, EM and TDDB become increasingly difficult with scaling. [13]

DRC and LVS are not optional final-stage clean-up tasks. The chosen PDK's rules should shape floorplanning from the beginning. In GF180, the public documentation states that designs submitted for fabrication must pass the foundry DRC checks, with formal waiver procedures required for accepted violations. [9] The same PDK imposes metal-density requirements; for example, its documented general metal rules include minimum density requirements that require dummy fill where necessary. [14]

SKY130 is useful for experimentation, but its public documentation warns that some collateral—particularly I/O infrastructure—is incomplete in the open release. [15] A commercial product should therefore use the actual foundry-supported sign-off decks and IP under the applicable agreement rather than assuming the open PDK contains everything required for production.

Manufacturability should drive several structural decisions: repeated tile shapes, regular SRAM placement, generous macro channels, redundant vias, conservative utilisation and uniform metal fill. GF's public DFM guidance explicitly recommends avoiding fragile minimum-rule geometries and, where possible, relaxing contact/via overlap beyond the minimum for manufacturability. [16]

DFT should include full scan on digital sequential logic, memory BIST for every SRAM instance, JTAG or equivalent test access, controllable clock/reset during test, scan compression if chain length requires it, and ATPG for stuck-at and transition faults. Four tiles should preferably use physically local scan partitions. MBIST should allow independent addressing of all five scratchpad-type SRAMs plus the instruction SRAM.

Yield improves if identical tiles use identical layout and SRAM orientations where possible. If tile memories become much larger in a later generation, spare rows or compiler-provided repair become worthwhile; at today's 1 KiB size, redundancy is unlikely to justify its cost unless the memory compiler already provides it.

Packaging is likely to determine the die size before the logic does. A low-pin-count version could use QFN or a small BGA. A true high-speed chiplet interface changes the problem substantially and would favour flip-chip BGA or advanced packaging because wire-bond inductance and limited edge bandwidth become constraints.

Process-node comparison

The process decision should be driven by required frequency, package, expected volume, verification maturity and product economics, not by the smallest available node.

Attribute	180/130 nm	65 nm	28 nm	7 nm
Device type	Planar CMOS	Planar CMOS	Advanced planar HKMG	FinFET
Current 100 MHz requirement	Very easy	Very easy	Extremely easy	Vast overkill
Relative logic area	1.0 baseline	~0.2–0.35	~0.07–0.15	~0.02–0.05
SRAM scaling	Moderate	Good	Very good	Excellent
I/O scaling	Poor	Poor/moderate	Limited	Limited
Likely die limiter	Pads + core	Pads	Pads / hard IP	Pads / hard IP
Dynamic energy	Highest	Lower	Much lower	Lowest
Leakage concern	Low/moderate	Moderate	Higher	High / process dependent
Routing difficulty	Low	Moderate	High	Very high
Sign-off complexity	Lowest	Moderate	High	Highest
IP availability	Excellent mature analogue	Strong	Strong	Strong but expensive
Mask/NRE burden	Lowest	Moderate	High	Extremely high
Best SilicaFlux use	Teaching / ultra-low-cost prototype	**First commercial ASIC**	Performance/ power revision	Only if major scale-up

The area factors above are **planning estimates**, not foundry guarantees. Geometric feature-size scaling alone substantially overestimates practical SoC area reduction because SRAM periphery, routing, clocking, analogue/IP and pads do not scale quadratically.

For 7 nm specifically, TSMC reports that N7 delivers up to around **3× logic density versus N16**, with up to 30% speed improvement or substantial power reduction depending on implementation choices. [17] Those are foundry platform comparisons, not a prediction that SilicaFlux itself would become exactly three times denser.

IRDS likewise warns that frequency scaling is increasingly limited by parasitics and thermal density rather than simply by transistor speed. Its 2024 roadmap explicitly notes that clock-frequency improvements become modest while wire parasitics and thermal limits become increasingly important. [18]

A rough SilicaFlux die projection is therefore:

Node	Estimated active core	Plausible packaged die target	Comment
180 nm	3–6 mm²	~9–16 mm²	Core and pads both material
130 nm	2–4 mm²	~6–12 mm²	Attractive open-PDK prototype
65 nm	0.8–1.5 mm²	~4–9 mm²	**Best first-product balance**
28 nm	0.3–0.7 mm²	~3–6 mm²	Mostly I/O limited
7 nm	0.1–0.3 mm²	likely >2–4 mm²	Hard IP / pad limited, poor economics

These are intentionally broad because no standard-cell synthesis, SRAM compiler or I/O count has yet been fixed.

Public 2026 Europractice pricing illustrates the cost discontinuity. Its 2026 schedule lists UMC 180 nm MPW access in the low-thousands of euros for specified shared blocks, while 65 nm runs are tens of thousands of euros under the published programme; advanced process pricing is increasingly quote-based. [19] Europractice and imec both explain that MPW combines multiple customers onto shared masks to reduce prototype cost. [20] TSMC's CyberShuttle service likewise states that shared prototyping can reduce mask-related prototype cost by as much as 90%. [21]

For planning rather than procurement, I would budget approximately:

Node	Prototype programme planning envelope*
180/130 nm MPW	£10k–£75k
65 nm MPW	£50k–£200k
28 nm MPW	£150k–£500k+
7 nm advanced-node prototype	£1m+ and potentially many millions

*These are **engineering budget envelopes**, not foundry quotes. They include the reality that fabrication is only part of total NRE; IP licensing, SRAM/PLL access, EDA sign-off, packaging, test development and engineering labour can exceed wafer cost. Public services often require direct quotes

for advanced nodes, so a specific 28 or 7 nm price should not be treated as known until a foundry programme provides it. TSMC and Europractice both operate shared-mask programmes specifically because mask NRE becomes material at advanced nodes. [22]

Concrete SilicaFlux implementation recommendation

The first fabricated SilicaFlux should be a **65 nm, single-digital-voltage, SRAM-macro-based ASIC organised as a symmetric 2×2 tile array**.

Use a square core. Put the scheduler at its centre. Place the four NoC routers immediately around it in a 2×2 arrangement. Place each tile SRAM directly outward from its router, then its vector engine and tensor unit outward from the SRAM. This ordering creates a radial structure:

scheduler → router → SRAM → datapath.

Place the scalar control core just west of the scheduler. Place the register bank between the decoder/control logic and scheduler. Place the instruction SRAM immediately north or west of fetch. Place the scalar scratchpad immediately adjacent to the pipeline MEM stage. The chiplet interface goes at the east or south edge. The PLL goes in a quiet corner on a dedicated analogue supply, separated from tile arithmetic.

A representative core could begin around **1.3 mm × 1.3 mm to 1.5 mm × 1.5 mm at 65 nm**, then be resized after synthesis and SRAM macro selection. Start at roughly 55–60% standard-cell utilisation. Maintain hard routing channels between the four tile regions and the centre. Do not place SRAM macros flush together; leave channels for PDN straps and their wide data buses.

The power plan should use a continuous VDD/VSS ring and orthogonal upper-metal straps across the four tile quadrants. Each tile receives at least two independent strap paths from the global grid. Add local decap near vector multipliers and SRAMs. The PLL receives separate filtered supply routing from dedicated pads or bumps.

Clocking should use one PLL or external clock receiver feeding a central root buffer close to the scheduler. From there use a balanced four-way top-level tree into one clock-gating point per tile plus a fifth branch for scalar/control logic. Tile-local CTS then distributes the clock internally. This topology is naturally symmetric, minimises skew and allows each idle tile to shut off most of its clock capacitance.

The highest-priority tapeout checklist is:

1. **Freeze one ASIC top-level configuration.** Fix `LANES`, SRAM depths, link width, external interfaces, clock target and reset convention; remove FPGA-only configurability from the sign-off configuration.
2. **Replace inferred memories with foundry SRAM wrappers.** Verify read latency, write masks, reset behaviour and MBIST access against the original RTL.
3. **Create ASIC constraints.** Replace the FPGA-oriented SDC assumptions with real input/output delays, PLL-generated clocks, PVT corners, clock uncertainty, false paths, multicycle paths and maximum transition/capacitance constraints.
4. **Close RTL quality before physical design.** Run lint, CDC, RDC, X-propagation, formal checks, constrained-random regression and coverage. The earlier assessment already identifies verification maturity as the project's main weakness. filecite turn0file0

5. **Run technology synthesis and obtain real area/power numbers.** The floorplan estimates in this report should then be replaced with library-derived data.
6. **Integrate DFT before floorplan freeze.** Scan, JTAG, test clocks, MBIST and SRAM repair signals materially change routing and pin planning.
7. **Floorplan hierarchically.** Preserve four physical tile partitions, centralise the scheduler, constrain SRAMs and reserve NoC channels.
8. **Build PDN before detailed placement.** Perform early static IR analysis, then dynamic IR with concurrent VMUL/DOT workloads.
9. **Run CTS with tile-level gating.** Close skew and insertion delay across all corners; avoid clock mesh unless frequency later demands it.
10. **Perform full sign-off.** STA with extracted parasitics, SI/crosstalk, IR/EM, DRC, LVS, antenna, density/fill, ESD/latch-up, thermal analysis and ageing.
11. **Validate manufacturability and package jointly.** Final pad ring, bump/pad spacing, power-pad count and package parasitics must be solved before GDS freeze, not afterwards.
12. **Tape out 65 nm first, then port to 28 nm only after silicon learning.** The logical architecture scales cleanly, whereas advanced-node physical complexity and NRE do not.

The central architectural insight is that SilicaFlux should **not** be laid out like a flattened CPU. Its best physical form is a regular accelerator fabric: four repeated compute islands, four local SRAMs, four routers and one compact central control/scheduling hub. That mapping preserves the RTL communication topology, shortens the highest-bandwidth local buses, creates a naturally balanced clock and power structure, distributes heat, simplifies hierarchical verification and gives the design a straightforward path from a conservative 65 nm first tapeout to a denser 28 nm implementation later.

---

[1] DCE Platform – N7/N6 Technology - Taiwan Semiconductor Manufacturing Company Limited
https://www.cld.tsmc.com/english/dedicatedFoundry/technology/platform_DCE_N7_N6?utm_source=chatgpt.com

[2] [5] [9] 2.0 Scope — GlobalFoundries GF180MCU PDK documentation
https://gf180mcu-pdk.readthedocs.io/en/latest/physical_verification/design_manual/drm_02.html?utm_source=chatgpt.com

[3] Hierarchical Macro Placement — OpenROAD documentation
https://openroad.readthedocs.io/en/latest/main/src/mpl/README.html?utm_source=chatgpt.com

[4] Power Distribution Network Generator — OpenROAD documentation
https://openroad.readthedocs.io/en/latest/main/src/pdn/README.html?utm_source=chatgpt.com

[6] Clock Tree Synthesis — OpenROAD documentation
https://openroad.readthedocs.io/en/latest/main/src/cts/README.html?utm_source=chatgpt.com

[7] A clock distribution network for microprocessors | IEEE Journals & Magazine | IEEE Xplore
https://ieeexplore.ieee.org/abstract/document/918917?utm_source=chatgpt.com

[8] Background — SkyWater SKY130 PDK documentation
https://skywater-pdk.readthedocs.io/en/main/rules/background.html?utm_source=chatgpt.com

[10] 7.14 Vian (where n = 1 to 5) — GlobalFoundries GF180MCU PDK documentation
https://gf180mcu-pdk.readthedocs.io/en/latest/physical_verification/design_manual/drm_07_15.html?utm_source=chatgpt.com

[11] [12] [13] [18] 2024 IRDS More Moore
https://irds.ieee.org/images/files/pdf/2024/2024IRDS_MM.pdf

[14] 7.13 Metaln (where n = 1 to 5) — GlobalFoundries GF180MCU PDK documentation
https://gf180mcu-pdk.readthedocs.io/en/latest/physical_verification/design_manual/drm_07_14.html?utm_source=chatgpt.com

15 Known Issues — SkyWater SKY130 PDK documentation
https://skywater-pdk.readthedocs.io/en/main/known_issues.html?utm_source=chatgpt.com

16 5.1 Guidelines for polygon shapes to be avoided — GlobalFoundries GF180MCU PDK documentation
https://gf180mcu-pdk.readthedocs.io/en/latest/physical_verification/design_manual/drm_05_1.html?utm_source=chatgpt.com

17 Smartphones Platform – Advanced Technologies - Taiwan Semiconductor Manufacturing Company Limited
https://www.cld.tsmc.com/english/dedicatedFoundry/technology/platform_smartphone_tech_advancedTech?utm_source=chatgpt.com

19 EUROPRACTICE | Schedules 2026
https://europractice-ic.com/schedules-prices-2026/?utm_source=chatgpt.com

20 EUROPRACTICE | MPW Fabrication
https://europractice-ic.com/services/fabrication/?utm_source=chatgpt.com

21 22 CyberShuttle® - Taiwan Semiconductor Manufacturing Company Limited
https://www.cld.tsmc.com/english/dedicatedFoundry/services/cyberShuttle?utm_source=chatgpt.com